NAME : UDDHAV BAPU GUND | DESIGNATION: JR.AWS DEVELOPER

TASK: VPC Peering Between Two VPCs

Create two VPCs and set up VPC Peering so that EC2 instances in each VPC can communicate with each other privately (without using the Internet).

Step:

Create Two VPCs

Launch EC2 Instances

Create VPC Peering Connection

Update Route Tables

Security Groups

Test Communication

Introduction

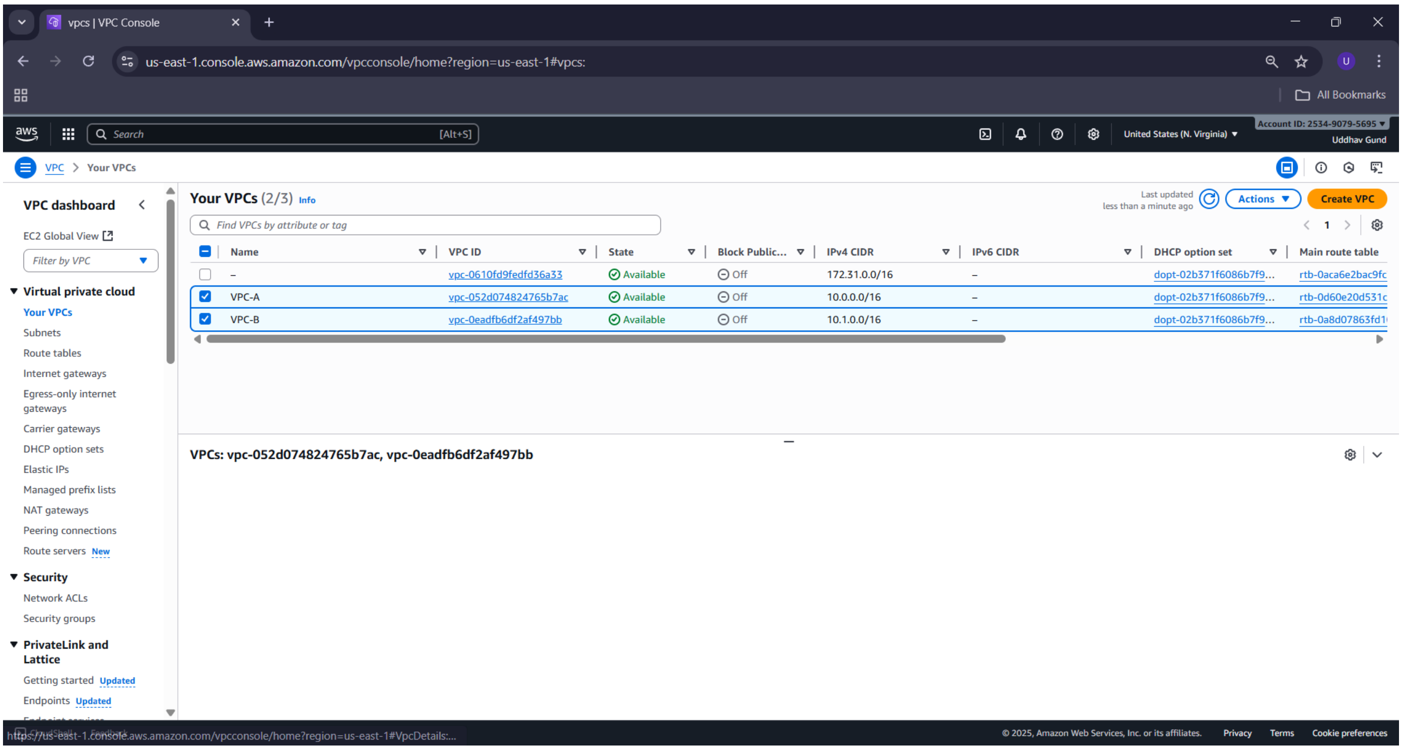
VPC Peering in AWS is a networking connection between two Virtual Private Clouds (VPCs) that enables instances in either VPC to communicate with each other using **private IP addresses**, as if they were part of the same network. This allows secure resource sharing without requiring internet gateways, VPNs, or external networks.

◆ Steps Executed

STEP 1 : CREATED TWO VPC's

VPC-A → CIDR 10.0.0.0/16

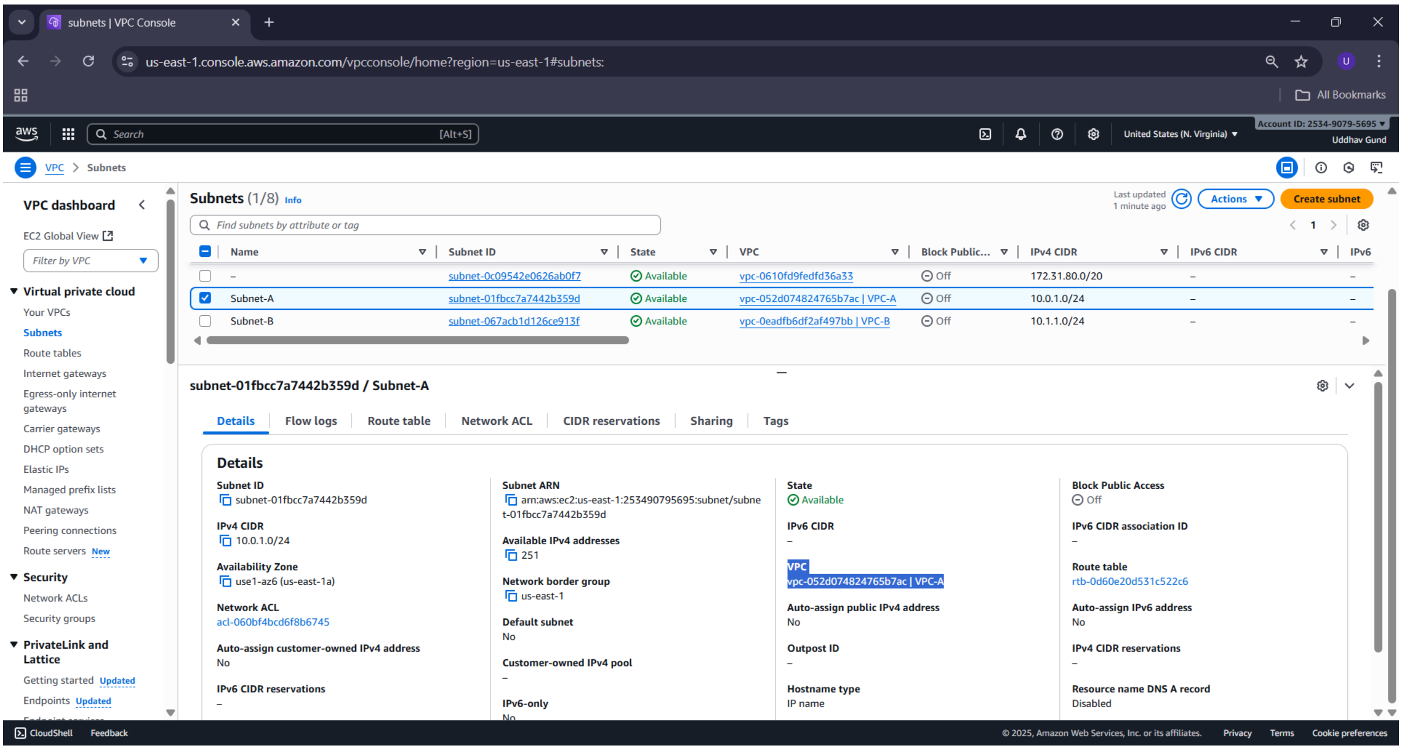
VPC-B → CIDR 10.1.0.0/16



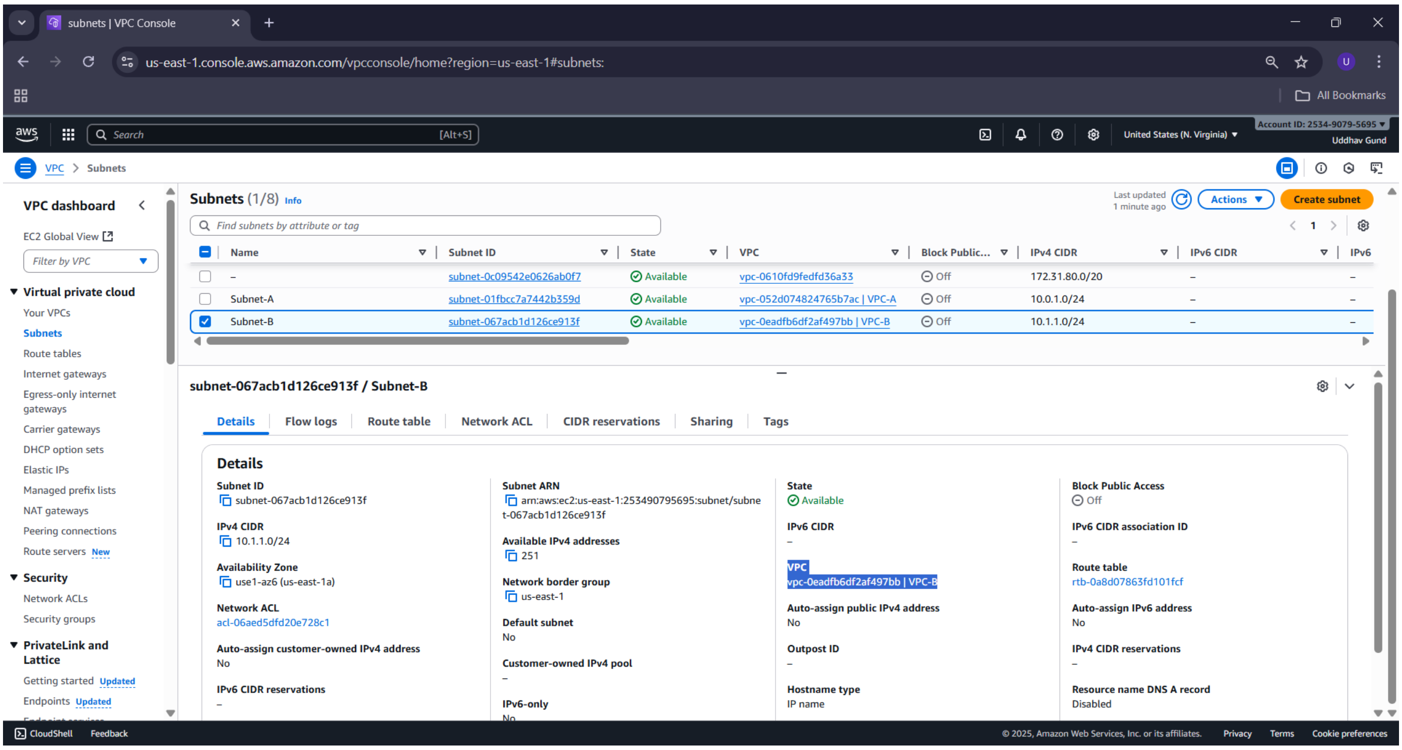
STEP 2: Created Subnets

- **Subnet-A → 10.0.1.0/24 in VPC-A**
- **Subnet-B → 10.1.1.0/24 in VPC-B**

Subnet-A

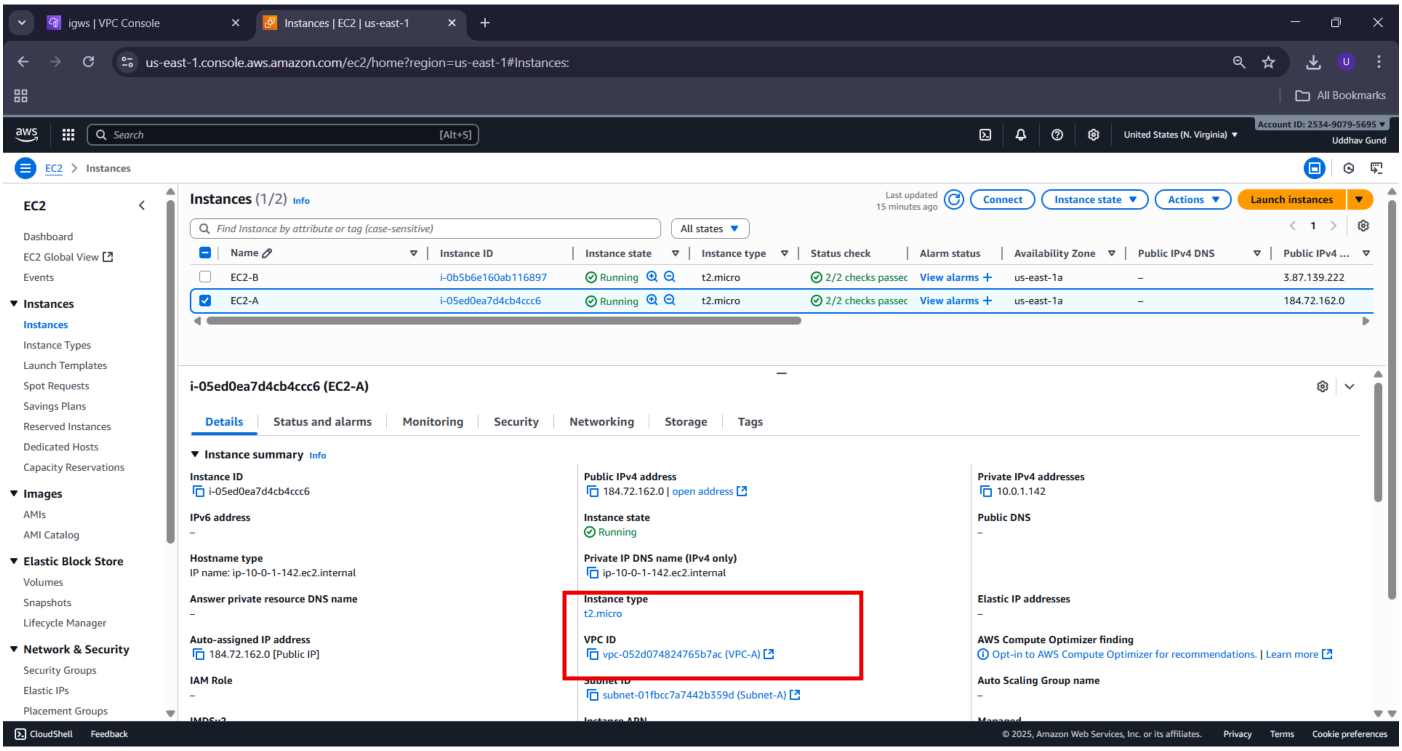


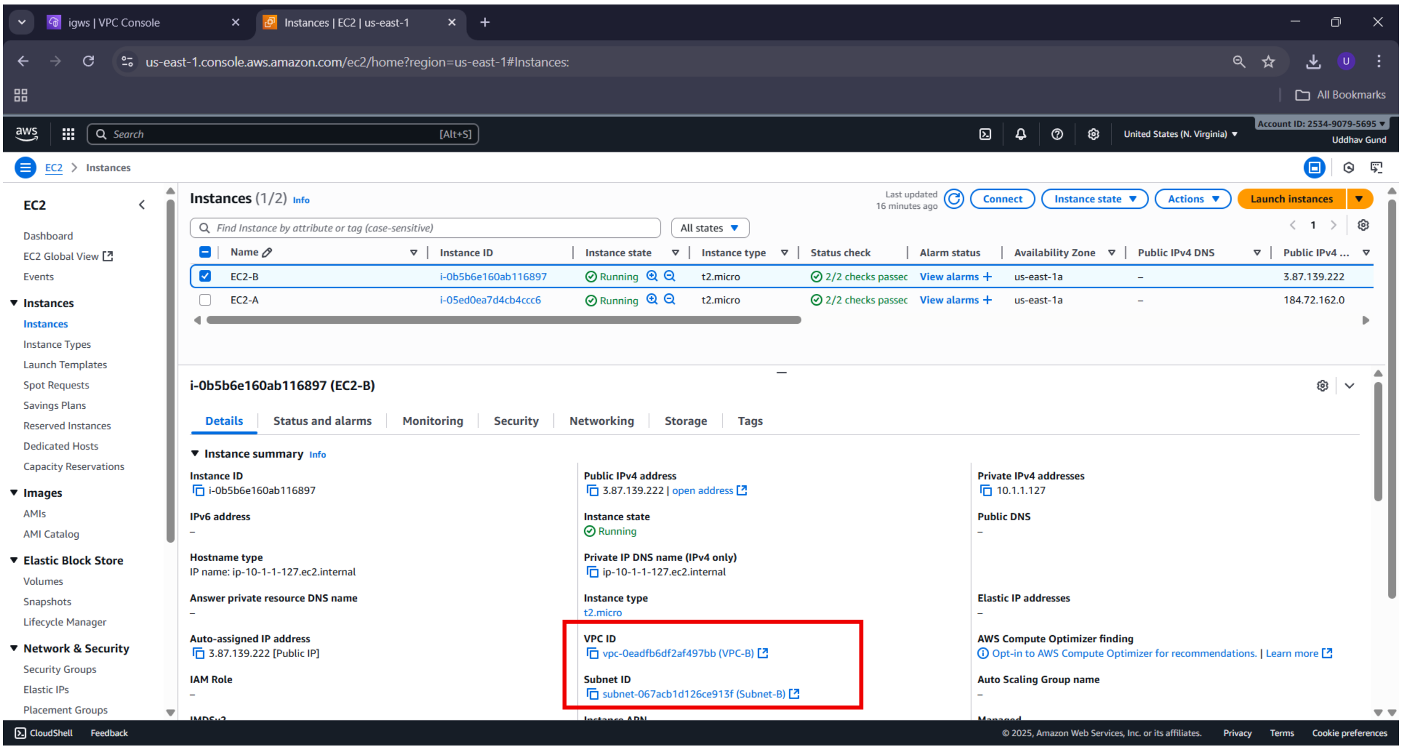
Subnet-B



STEP 3: Launched EC2 Instances

- **EC2-A in VPC-A → Private IP 10.0.1.142**
- **EC2-B in VPC-B → Private IP 10.1.1.127**

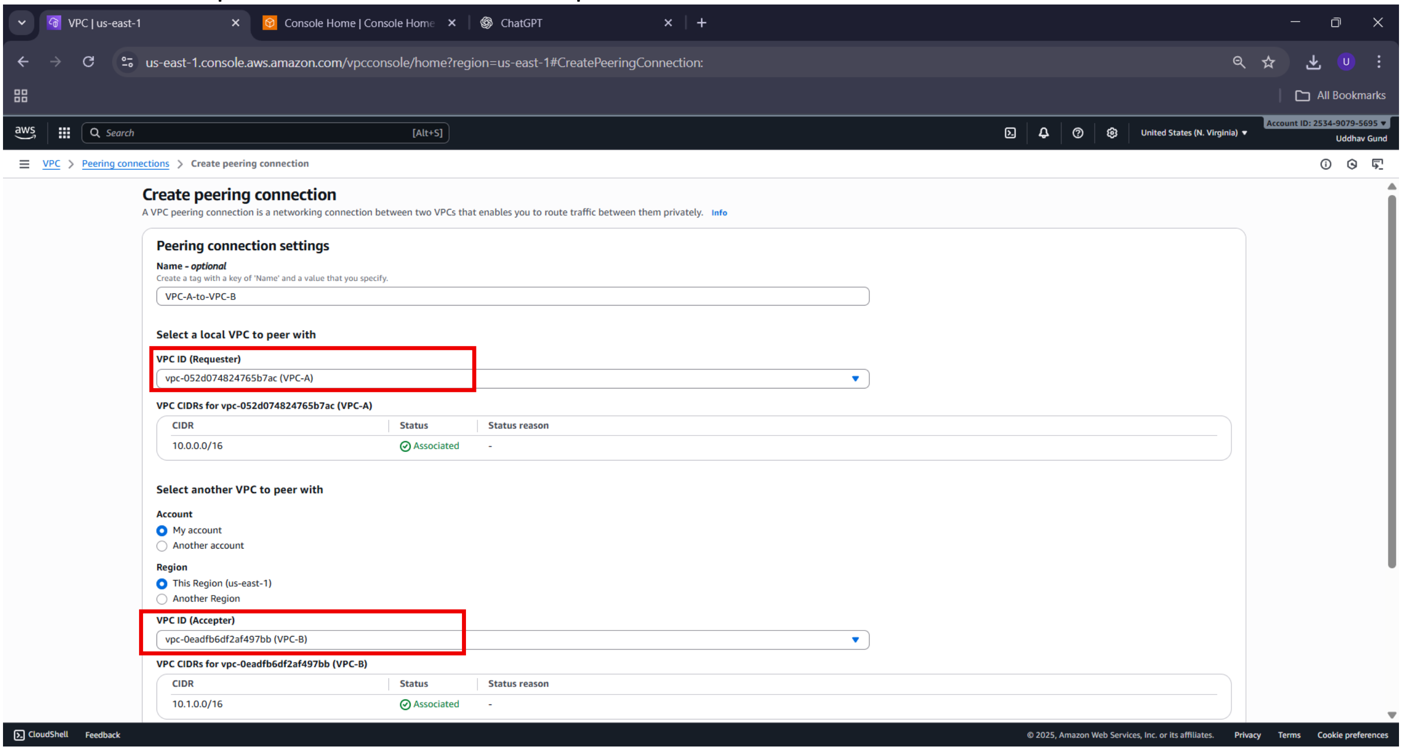


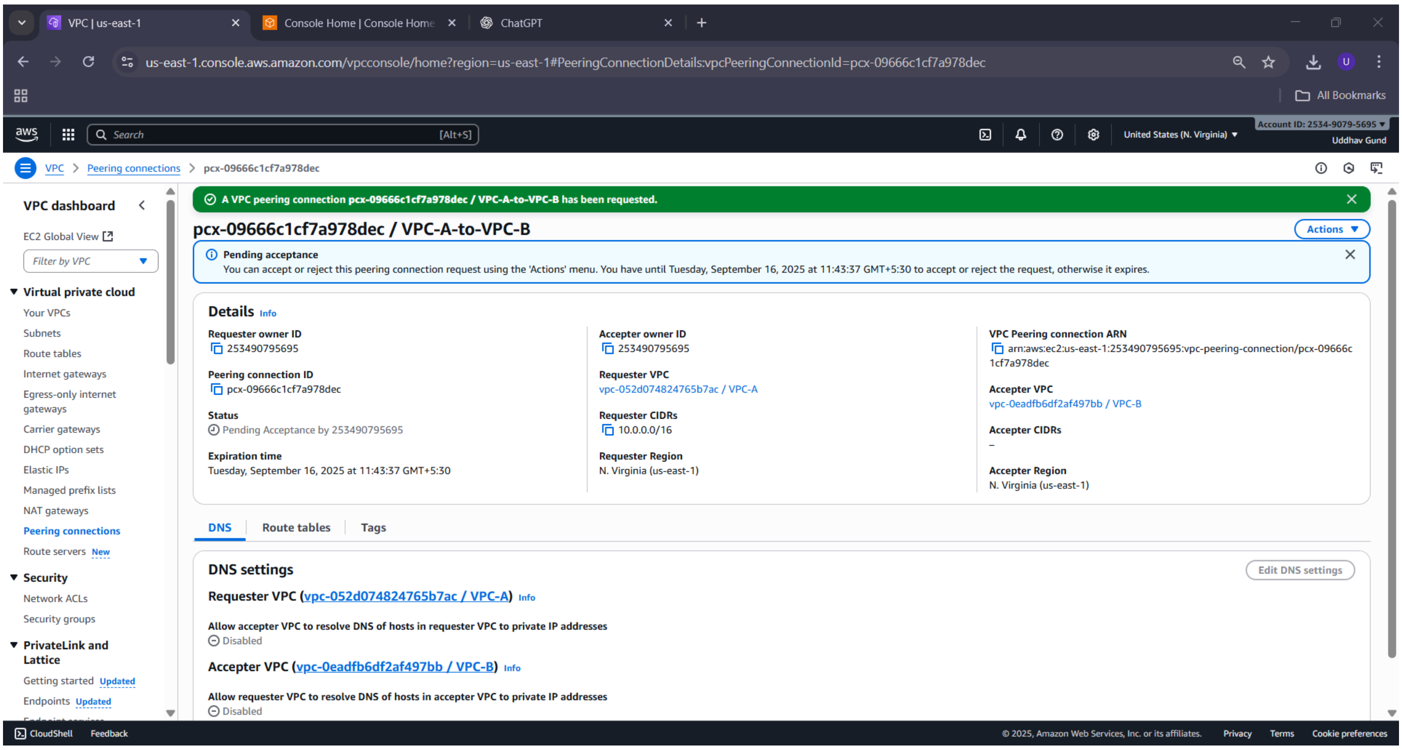


STEP 4: Created VPC Peering

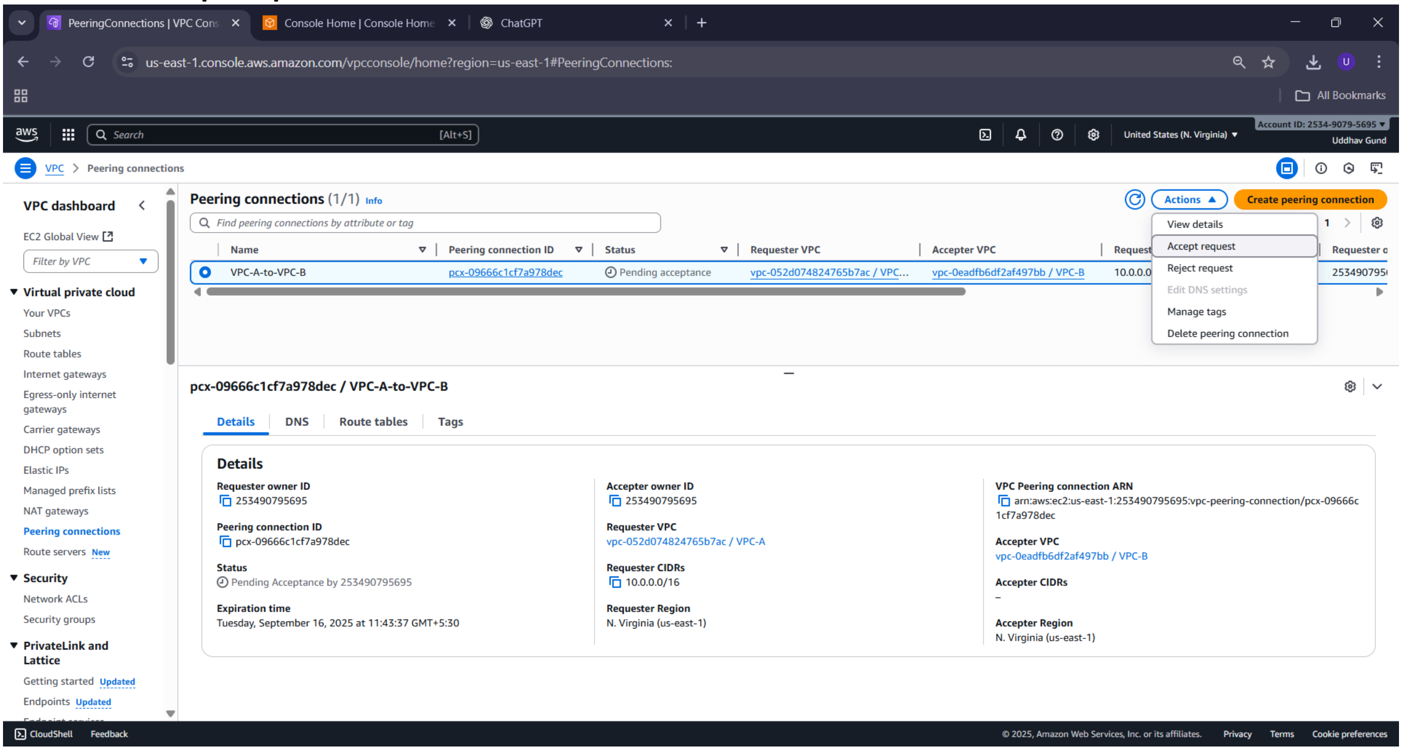
- **Peering established between VPC-A and VPC-B**
- **Accepted successfully**

Set VPC-A as Requester and VPC-B as a Accepter





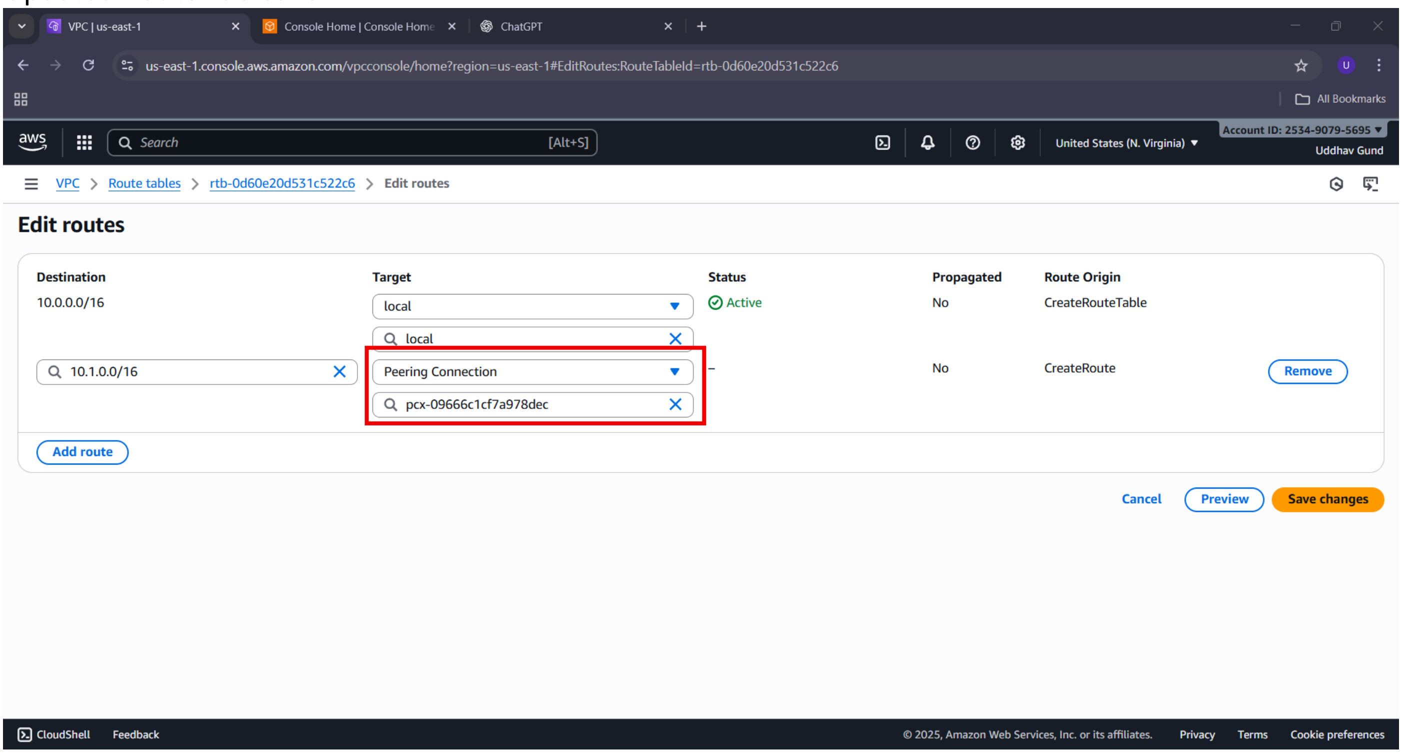
Actions →Accept request for make status in Active mode



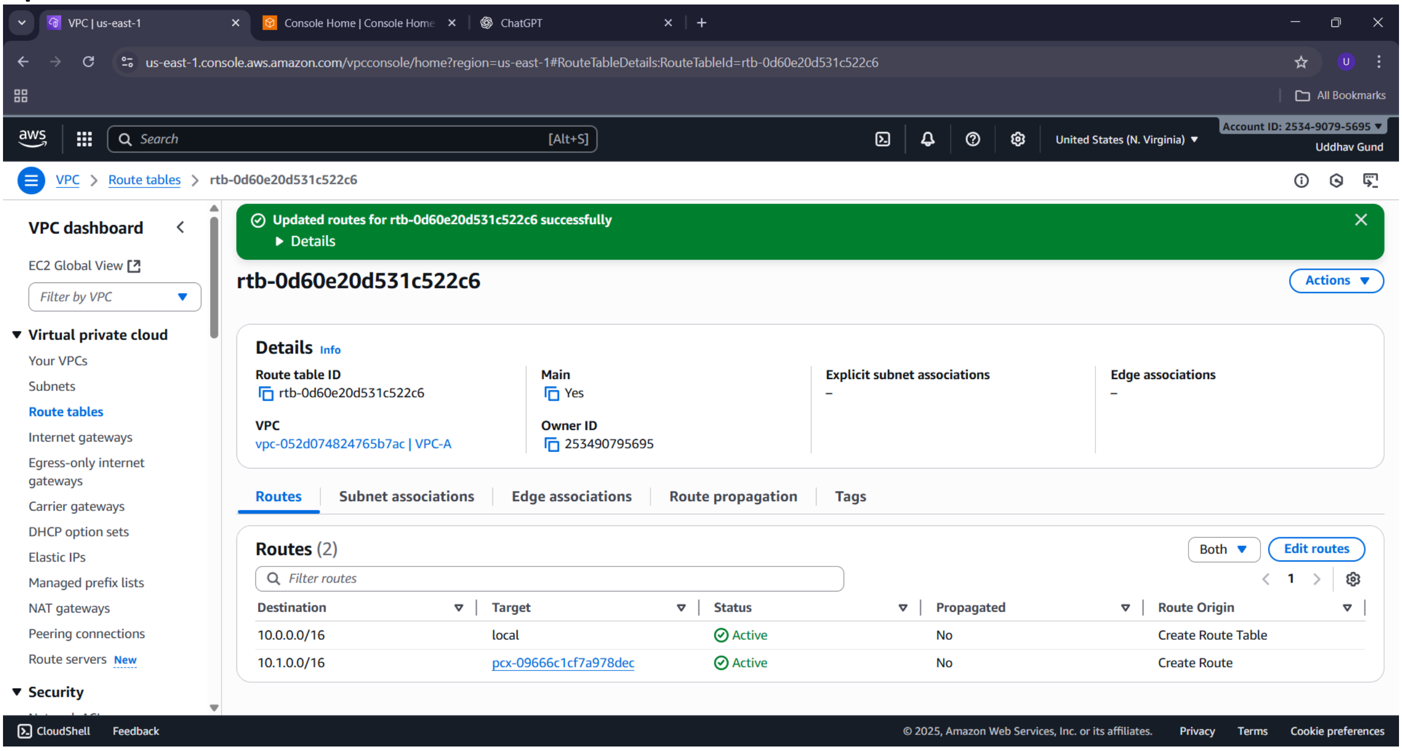
STEP 5: Updated Route Tables

- **In VPC-A → route 10.1.0.0/16 → Peering Connection**
- **In VPC-B → route 10.0.0.0/16 → Peering Connection**

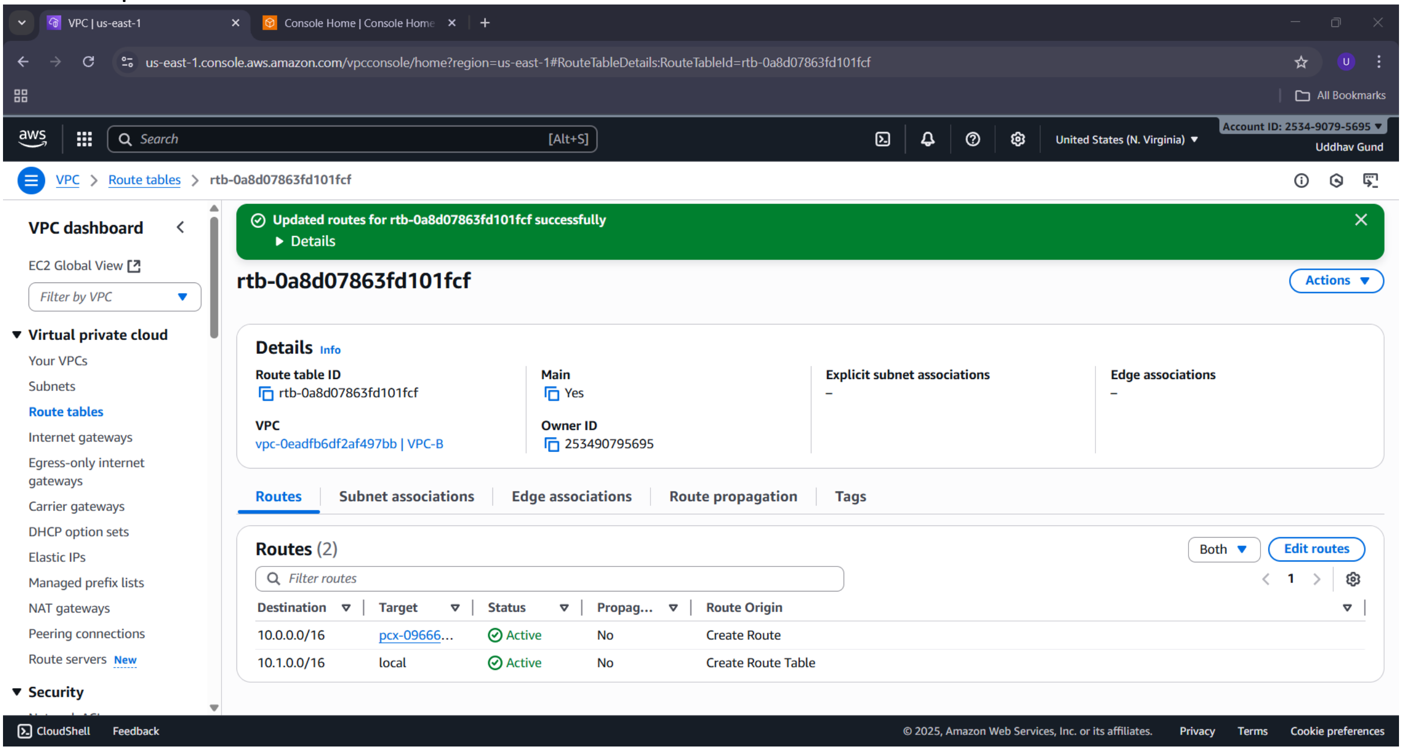
Updated Route Tables for VPC-A



Updated!



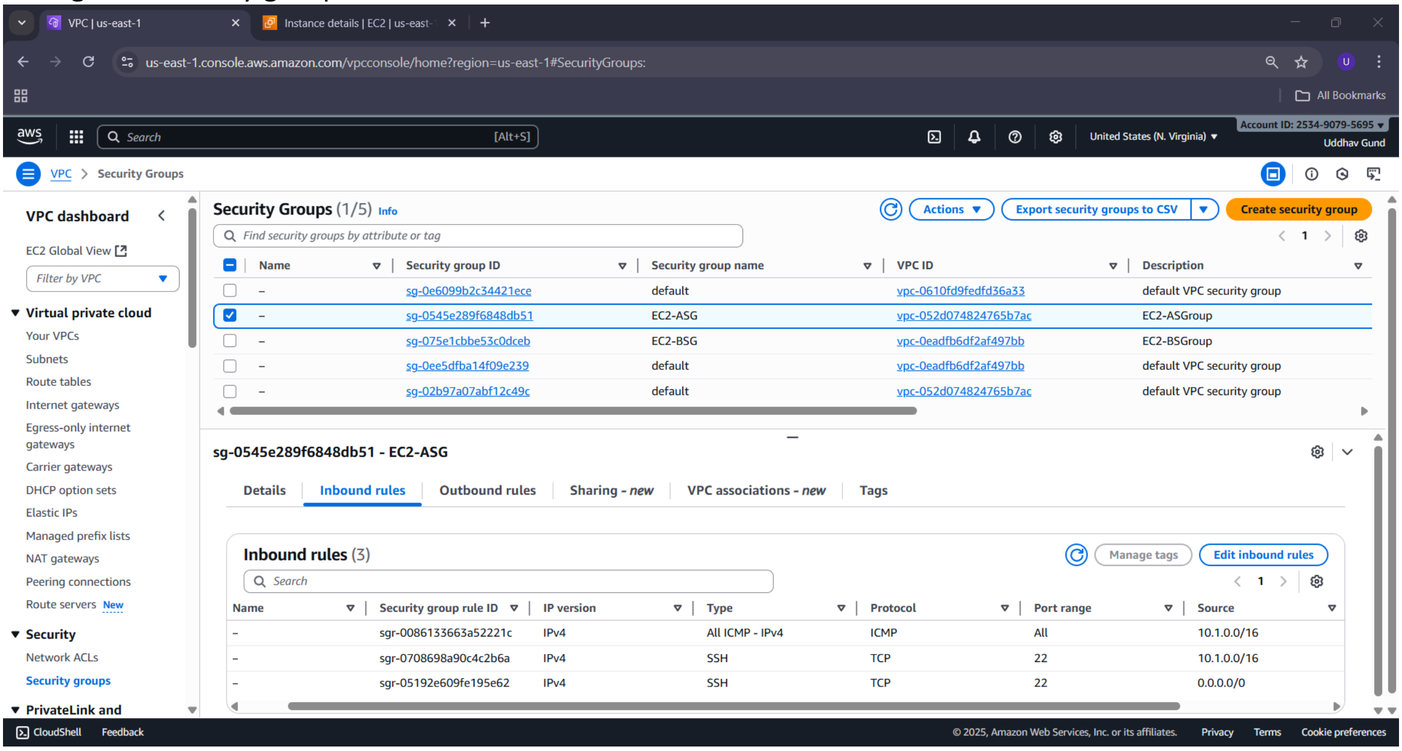
Same step for VPC-B



STEP 6 : Configured Security Groups

- **Allowed ICMP (ping) and SSH (22) within VPC CIDRs**
- **Allowed HTTPS (443) for SSM Endpoints**

Configured security groups EC2-ASG of VPC-A and EC2-BSG of VPC-B



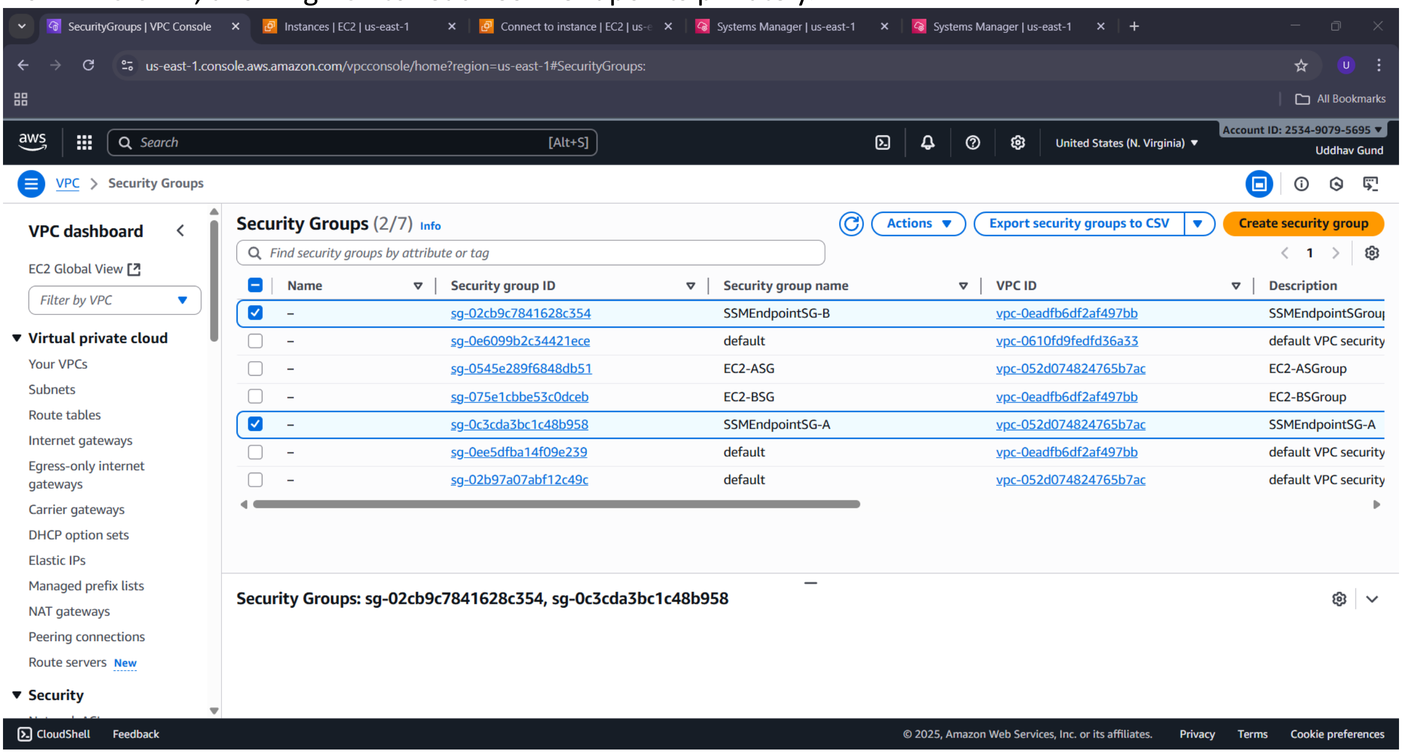
STEP 7 : Created VPC Endpoints

- **Added Interface Endpoints for ssm, ec2messages, ssmmessages in both VPCs**

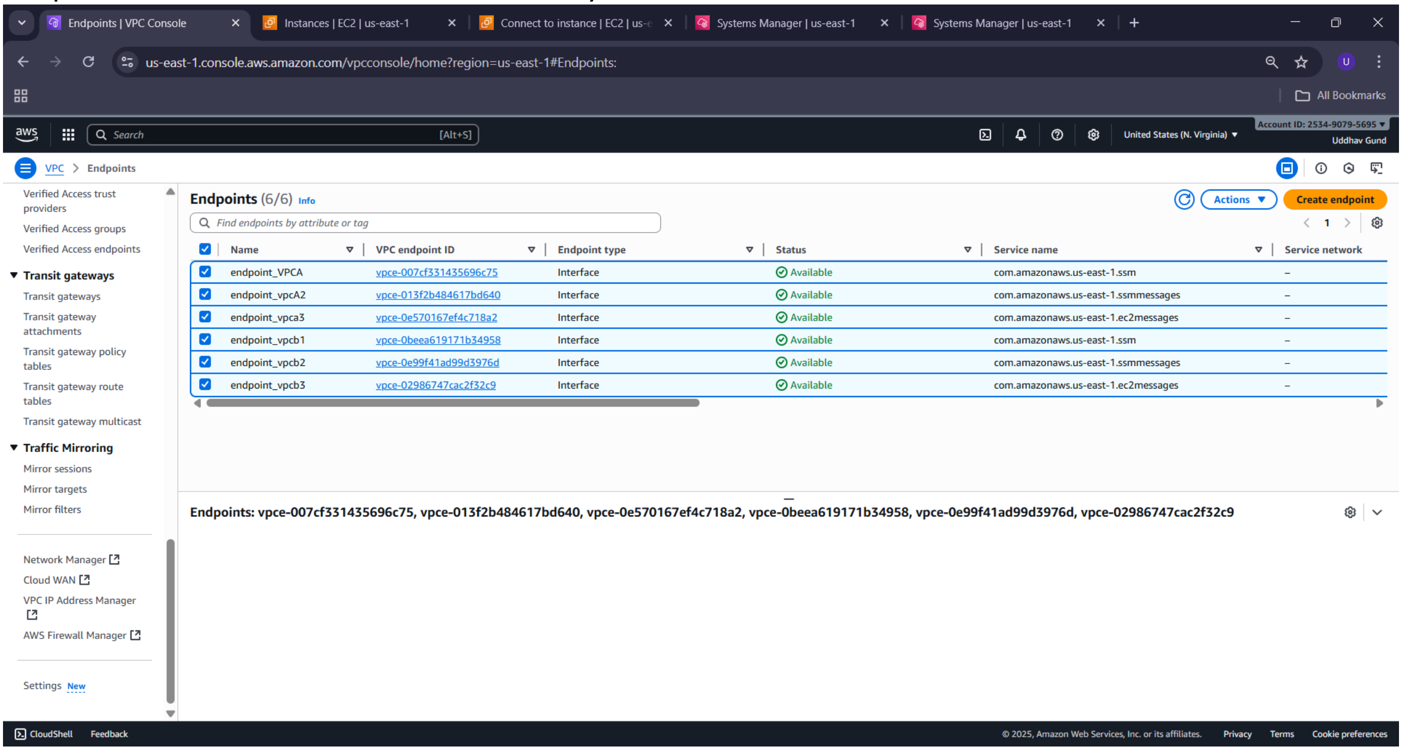
[**Why Endpoints & Session Manager?**

We used **Session Manager** to access EC2 without public IP/SSH, and **VPC Endpoints (ssm, ec2messages, ssmmessages)** so the SSM Agent could communicate privately. This design ensures compliance with the requirement of *private communication without internet*.]

Endpoint Security Groups – Created **SSMEndpointSG-A** & **SSMEndpointSG-B** with inbound HTTPS (443) from VPC CIDR, allowing EC2 to reach SSM endpoints privately.

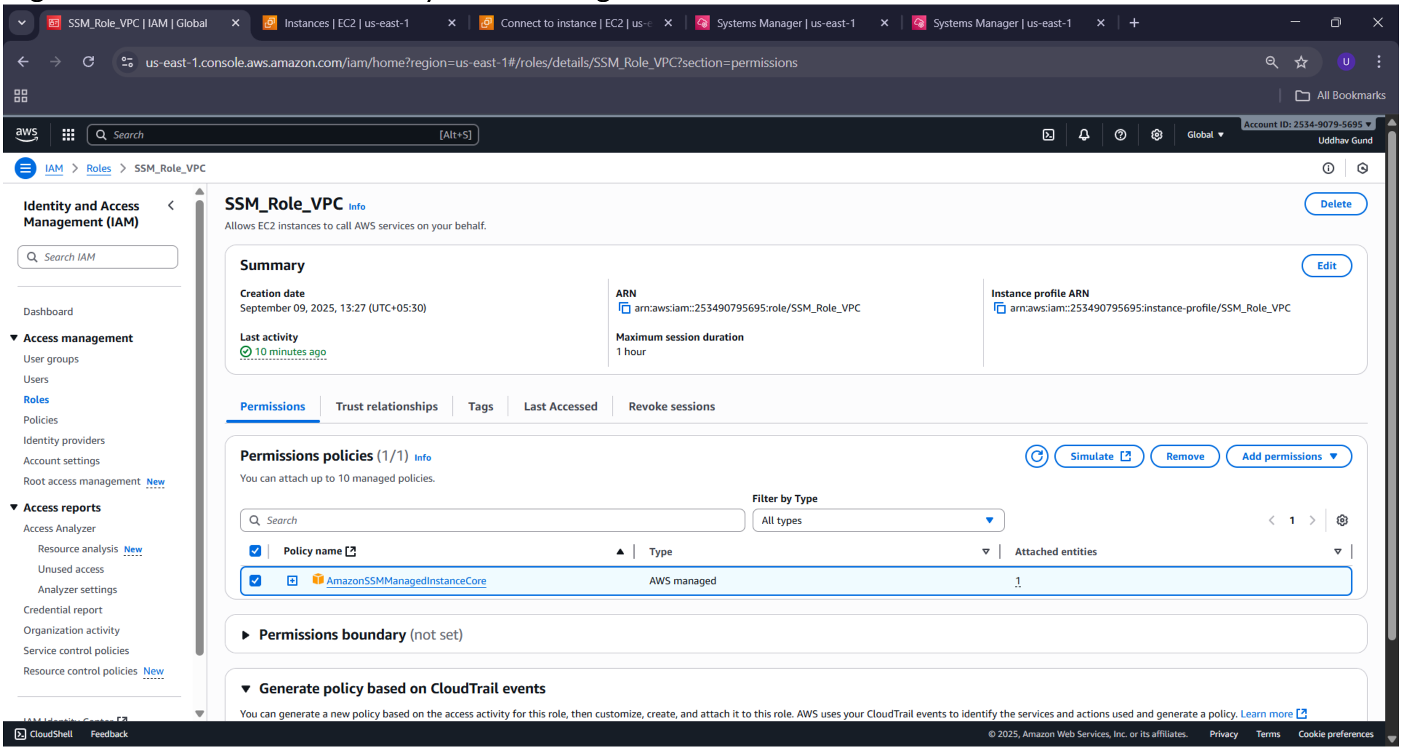


Endpoints of VPC-A and VPC-B simultaneously!



IAM Role for Session Manager

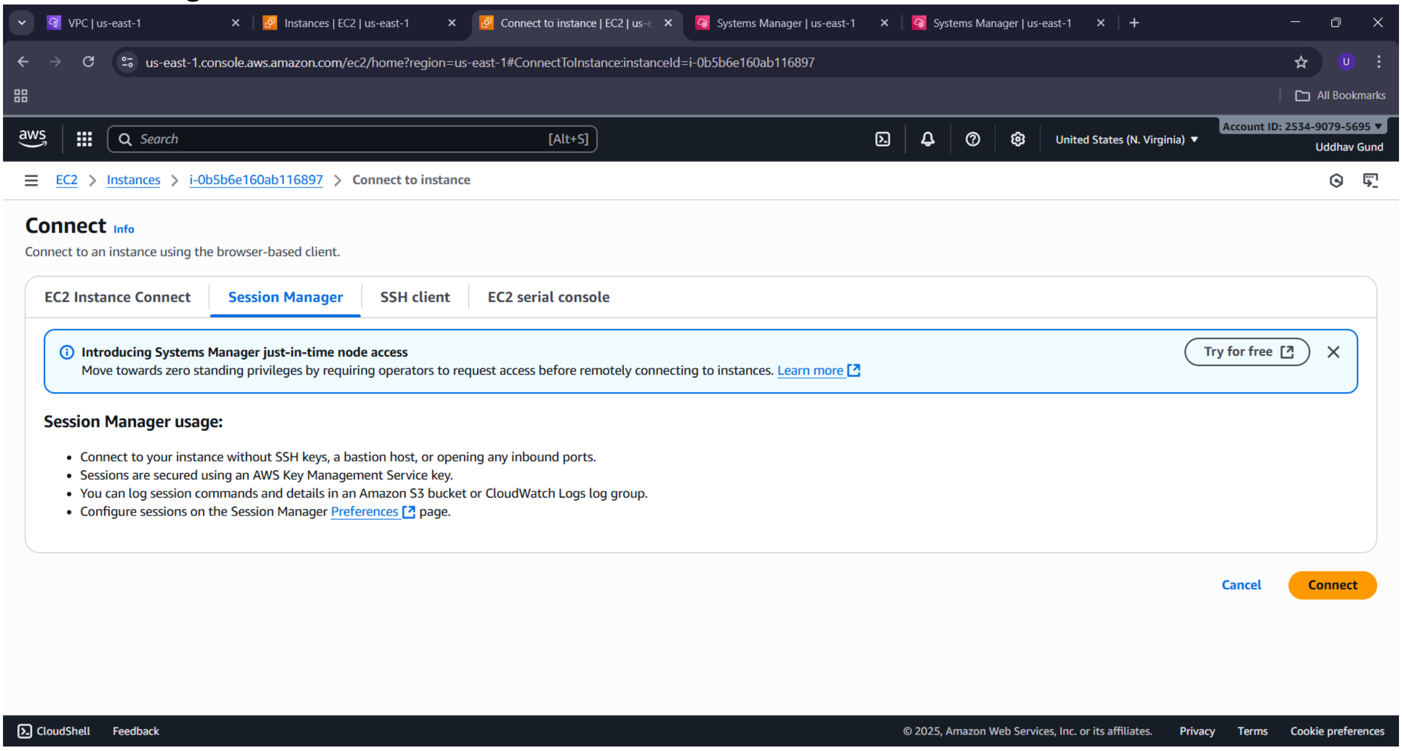
Attached IAM role with **AmazonSSMManagedInstanceCore** policy to EC2s, enabling the SSM Agent to register and connect with AWS Systems Manager.



STEP 8: Verified Tested Private Connectivity

1. Used Session Manager for access
2. Ping test from EC2-A → EC2-B ☑
3. Ping test from EC2-B → EC2-A ☑

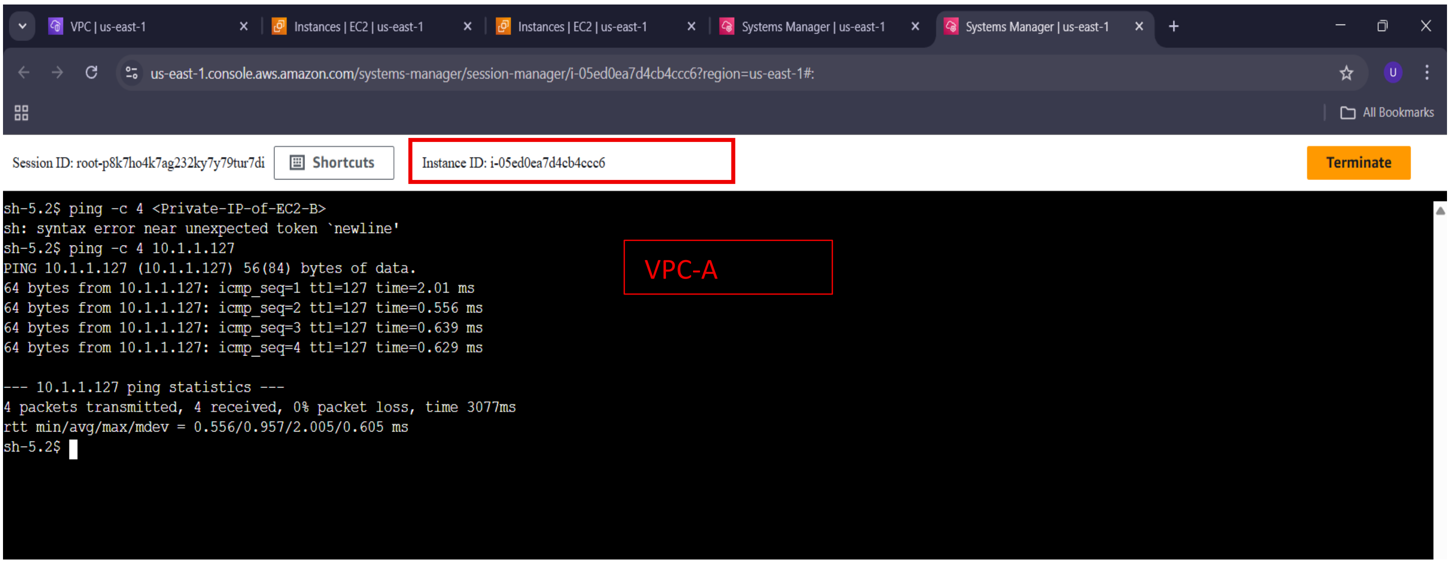
Session Manager!



- **On EC2-A shell, ping EC2-B private IP:**

ping -c 4 <EC2-B private IP>

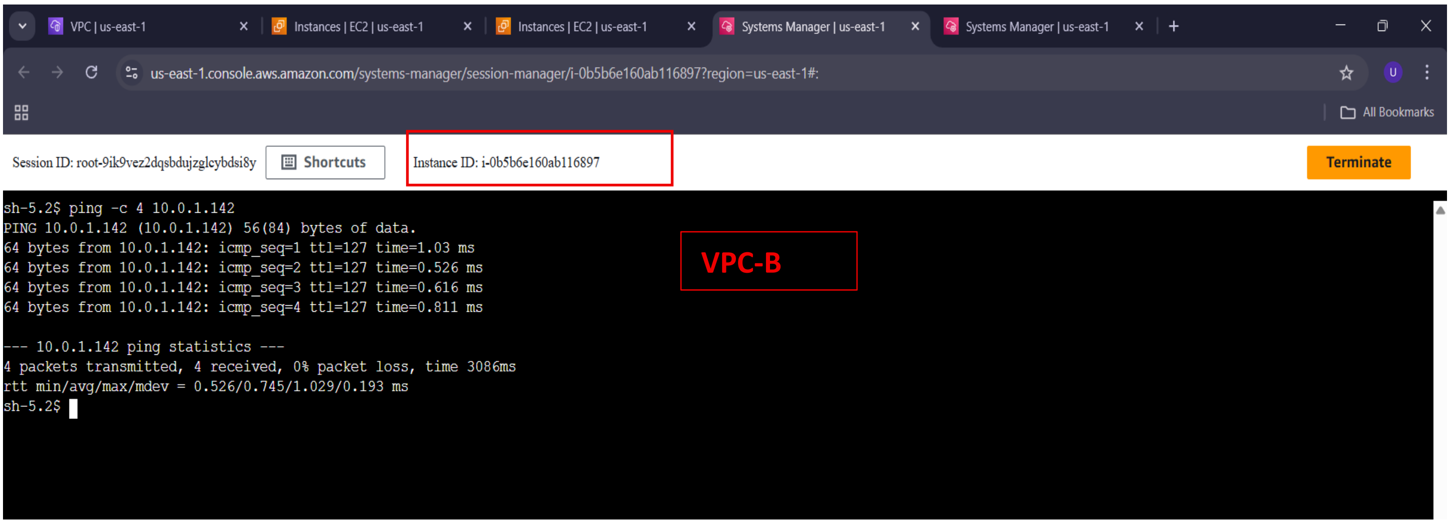
From **EC2-A (10.0.1.142):** ping -c 4 10.1.1.127



- On **EC2-B shell**, ping EC2-A private IP:

ping -c 4 <EC2-A private IP>

From **EC2-B (10.1.1.127):** ping -c 4 10.0.1.142



✅ Conclusion :

- VPC Peering established successfully.
- Endpoints + Session Manager enabled **private management without internet**.
- EC2 instances in different VPCs communicated using **only private IPs**.
- Task completed as per requirements.